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Multi-ancestry finemapping of gene clusters implicated in cardiovascular disease and mental health

A genomic region containing a cluster of genes that regulate fatty acid desaturation has recently been found to have a strong association with major depressive disorder (Adams et al. 2025). However, it is unclear which variant in this region is the cause of the association. This project used biobank whole genome-sequencing datasets to finemap common variants, catalogue rare variants, and understand shared genetic risk between cardiovascular disease and common psychiatric disorders.

The FADS gene cluster

The fatty acid desaturase gene family (FADS1, FADS2, FADS3) is co-located in a gene-dense region on chromosome 11 (11q12-q13.1). FADS1 and FADS2 regulate the conversion of polyunsaturated fatty acids (omega-6 and -3) into long-chain forms, which have roles in a wide range of cellular processes and are implicated in cardiovascular, metabolic, and immunological health (Koletzko et al. 2019).

The role of FADS as the genes underlying the association with major depression is bolstered by a genetic causal analysis

(Mendelian randomisation) of circulating metabolites (Davyson et al. 2023), which showed that omega-3 levels and omega-6 to omega-3 ratios had a causal effect on depression. While this makes FADS1 and FADS2 the most plausible biological candidates, expression levels of other genes in this region (MYRF and TMEM258) also have possible links to depression. Further work is therefore required to pinpoint the causal variant driving the association.

Genetic associations

Prior studies examining specific FADS genetic variants and depression (Sublette et al. 2016; Cribb et al. 2018) had small sample sizes ($N \approx 200$ –1000). Large genome-wide association studies (GWASs) have identified a strong association signal in this region, but the most significant variant has been located in other nearby genes like DAGLA (Howard et al. 2019; Meng et al. 2024) and TMEM258 (Levey et al. 2021; Als et al. 2023). However, the most recent and largest multi-ancestry genetic study of depression found the top signal to be a single nucleotide intron variant in FADS2 (Figure 1).

Finemapping

Because genetic variants in a region are in close physical linkage, association studies find a correlated statistical signal across variants that makes it difficult to determine which variant is the cause of the association. Finemapping is the process of comparing the linkage and association signals to determine the probability that each SNP is the causal SNP. Finemapping can be enhanced by considering prior causal probabilities based on functional annotations (such as a coding or regulatory function, or evidence of mutational constraint).

Previous investigations of this region used (1) association summary statistics from European-ancestry cohorts only and (2) summary statistics from all ancestry groups merged together. Pre-computed information from UK Biobank was used and it was assumed that the region only had a single causal variant.

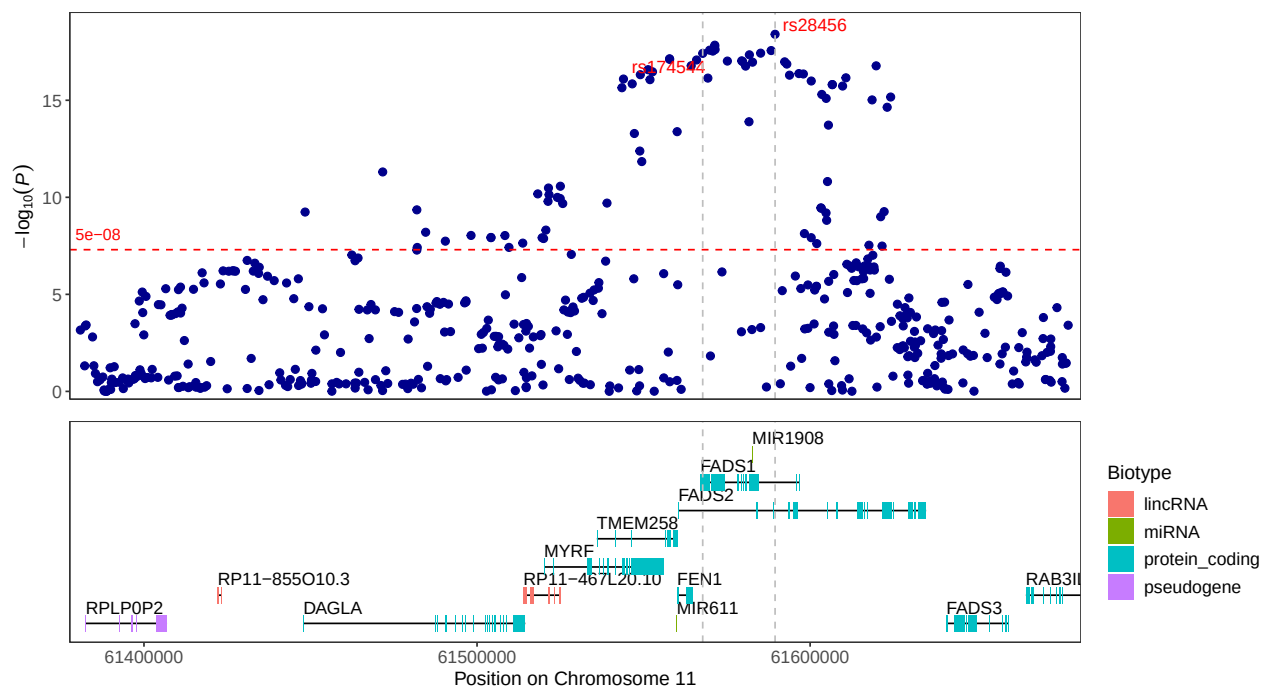


Figure 1: Region plot of genome-wide association findings for the FADS gene cluster. Top: Tested variants with transformed association p-value. Bottom: Gene positions in the region.

This latter assumption was stipulated because it does not require that the linkage disequilibrium estimates be population accurate. The assumption of a single causal variant was also reasonable given that the conditional-and-joint analysis (which tests whether each other variant is still significant after controlling for the top associated variant in the region) did not indicate the presence of multiple causal variants.

However, this analysis did not identify a variant that met the finemapping identification criterion, which was a posterior inclusion probability (PIP) > 0.95 . The most significant variant in the multi-ancestry analysis was rs28456¹ while the finemapping analysis selected rs174544² as the more likely causal SNP.

¹ [rs28456](#): an intron variant in the [FADS2](#) gene and the uncharacterised gene [LOC124902680](#); PIP = 0.122.

² [rs174544](#): a variant in 3' untranslated region of [FADS1](#); PIP = 0.181.

Rare variant burden

An analysis to look for the effect of rare variants on major depression and on metabolite levels has also been conducted (Nisbet et al. 2025). This study used short read whole genome sequencing (srWGS) datasets from All of Us and UK Biobank. It considered metabolic measures that had been associated with major depression (Omega 3, Omega 3 percentage, Docosahexaenoic acid, Omega 6/Omega 3 ratio and Degree of Unsaturation.). The authors found that rare variant burden was significantly associated with all tested metabolite measures but was not associated with major depression. This indicates that the GWAS association is likely to be caused by a common variant than being a proxy for a rare variant.

Analyses

In this project I aimed to discover more about causal variants in the FADS region using multi-ancestry and whole-genome sequencing datasets. Initial analysis improved on previous methods in three ways:

- Rather than using summary statistics from merged ancestries, use summary statistics from the single largest

ancestry group with a matched reference panel to test for multiple causal variants.

- Use disaggregated summary statistics from multiple ancestries to improve finemapping.
- Use a large, multi-ancestry-matched WGS dataset as an improved reference panel for finemapping.

All analyses were focused on the FADS gene region `chr11:61400000-61700000` using summary statistics from Adams (2025) and Meng (2024). The ancestry clusters used were derived from continental reference populations, namely African (AFR), Admixed American/Hispanic/Latino (AMR), East Asian (EAS), European (EUR), and South Asian (SAS). I conducted functionally-informed finemapping using PolyFun (Weissbrod et al. 2020) and multi-ancestry finemapping with SuSiEx (Yuan et al. 2024). I looked up variant functional annotations using the myvariant package (Mark 2024).

Finemapping attempts to identify “credible causal sets”, which are a group of variants that collectively capture an association signal in a region of the genome. The variants are ranked by the a posterior inclusion probability (PIP), which combines the prior functional probabilities with the data to determine a posterior causal probability for each variant. A $PIP > 0.95$ is typically taken as sufficient to identify the causal SNP, but values less than this are still useful for narrowing down the set of potentially causal variants.

Test for multiple causal variants

Testing for multiple causal variants requires an ancestry-matched reference panel. I started with the standard reference panel from the 1000 Genomes Project (The 1000 Genomes Project Consortium 2015). I used prior probability information from PolyFun (Weissbrod et al. 2020) and then performed finemapping using FINEMAP (Benner et al. 2016) on the European-ancestry cluster association summary statistics ($N_{\text{cases}} = 525,197$; $N_{\text{controls}} = 3,362,335$).

The finemapping analysis found three credible causal sets which contained 6, 2, and 175 variants respectively (Table 1). While

no SNP had a PIP > 0.95 , this analysis supports the conclusion that there are multiple causal variants in the FADS region. The third credible causal set (CCS) contained the SNP (rs174544) that was the top finemapped variant in the published analysis while also containing MYRF³ intron variants. The first CCS contained SNPs that were primarily in FADS2 introns, but there was one variant (rs174567) that was an upstream transcript variant. The second CCS contained intron variants in DAGLA.⁴

Table 1: Top five SNPs with highest posterior inclusion probability (PIP) in each credible causal set from finemapping of European-ancestry cluster with 1000 Genomes reference panel.

Credible set	SNP	Position (b37)	PIP
1	rs174567	61593005	0.006
1	rs99780	61596633	0.148
1	rs174574	61600342	0.037
1	rs174580	61606642	0.006
1	rs174583	61609750	0.134
2	rs198433	61484576	0.073
2	rs198449	61498197	0.018
3	rs12805732	61448384	0.336
3	rs198465	61521318	0.103
3	rs198462	61524119	0.167
3	rs174544	61567753	0.212
3	rs174561	61582708	0.268

³ MYRF: myelin regulatory factor.

⁴ DAGLA: diacylglycerol lipase alpha, involved in biosynthesis of endocannabinoid ligands.

Improving the reference dataset

While standard reference samples like 1000 Genomes (The 1000 Genomes Project Consortium 2015) are convenient for downstream analysis, the small panel sizes are suboptimal for finemapping (Benner et al. 2017). To improve on this, I used a much larger whole genome sequencing dataset from the All of Us cohort (Bick et al. 2024). I used the short read whole genome sequencing (srWGS) dataset version 8 that had been filtered for allele frequency and allele count, which matches the complement of variants typically found in a GWAS. I matched

the MDD summary statistics to each ancestry cluster and randomly extracted up to 10,000 participants from each cluster to compose the linkage disequilibrium reference panel.

The improved reference dataset identified 1–4 credible causal sets for each ancestry cluster (Table 2). The highest PIP (= 0.718) was for rs7115739⁵ in the EAS cluster. While still not conclusive, the findings had higher posterior causal probabilities compared to the standard reference panel analysis (for example, PIP for rs12805732 in the EUR cluster 0.336 → 0.446).

⁵ rs7115739, intron variant in FADS3.

Table 2: Top SNP with highest posterior inclusion probability (PIP) in each credible causal set from finemapping of each ancestry clusters using All of Us WGS as the reference panel.

Cluster	Credible set	SNP	Position (b38)	PIP
AFR	1	rs198456	61695916	0.275
AFR	2	rs17827918	61700693	0.168
AMR	1	rs57398279	61971670	0.103
AMR	2	rs17156616	61975806	0.591
AMR	3	rs115430155	61984156	0.640
EAS	1	rs2727260	61944770	0.574
EAS	2	rs7115739	61874245	0.718
EAS	3	rs174455	61888645	0.079
EAS	4	rs174464	61890454	0.147
EUR	1	rs174583	61842278	0.265
EUR	2	rs198433	61717104	0.074
EUR	3	rs12805732	61680912	0.446
SAS	1	rs498793	61857233	0.069

Multi-ancestry finemapping

Finally, I conducted a multi-ancestry meta-analysis with the All of Us srWGS datasets as a reference panel. The analysis identified only one credible causal set (Table 3). The putative SNPs were all introns or upstream variants for FADS1 and FADS2.

Table 3: SNPs in credible set from joint finemapping of all ancestry clusters using All of Us WGS as the reference panel.

Credible set	SNP	Position (b38)	PIP
1	rs174544	61800281	0.025
1	rs174548	61803876	0.327
1	rs174549	61803910	0.082
1	rs174560	61814292	0.102
1	rs28456	61822009	0.424

Conclusions

Using a multi-ancestry sequencing dataset provided a improvement for finemapping discovery in a complex genetic region. The results advanced over published analysis (Adams et al. 2025) by showing that there is likely more than one causal variant in the region and suggesting that causal effects possibly differ between ancestry clusters. While the evidence for pinning down an exact causal variant was marginally improved, it confirms that the FADS region contains a complex and interesting gene set to investigate.

Project management and researcher development

Support for this project has enabled data access, compute, and storage of the All of Us dataset using the cloud-based Researcher Workbench. I have focused on the All of Us cohort because it has a larger sample of non-European ancestry groups and a version of the whole genome sequencing dataset that is highly amenable to finemapping genome-wide association results.

The project was initially held back by organising billing set up to buy credits for the cloud computing platform. The initial plan was to set up invoicing through a purchase order. However, the scope of this project was below the threshold cloud providers would accept to create an invoicing account. There

might be a need for a centre- or institute-level billing arrangement that allows smaller projects to group together to satisfy billing preferences.

This project has been extremely beneficial to my scientific and career development. It allowed me to develop skills in two areas (finemapping and handling whole genome sequencing datasets). The experience has contributed in multiple ways. It has helped me feel confident in leading future work in this area. The knowledge I gained helped me write a workstream based on the ideas and datasets that I have been able to explore. It also put me in a better position to supervise a research student who is also working with these sequencing datasets.

Acknowledgments

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